1. Title of the project

Agent-Based Modelling (ABM) predicts number needed to vaccinate to achieve a 50% Reduction in Zero-Dose Vaccination among Under-five Children in Kenya by 2025

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2. Abstract

Introduction: Zero-dose proportion remains a challenge in Kenya, with 7% of infants affected, according to Kenya Demographic Health Survey 2022. One in five children lack access to vaccines in **Sub Saharan Africa**. Globally, 14.2% are penta-zero-dose with 7.5% truly zero dose, indicating a lack of critical vaccinations among under-fives. Achieving WHO global target of 90% immunization coverage is vital to prevent 2.5 million vaccine-related deaths annually, among under-fives.

Methodology: Agent based modelling was employed to project the reduction needed in zero-dose vaccination rates to achieve 50% decrease by 2025 in Kenya. Currently: diphtheria is eliminated; neonatal tetanus is near elimination; pertussis and measles have marked control; therefore, tetanus was adopted in the model. Agents alive and zero-dose were included in fitting the model. ***Starsim- SIS model*** was employed with tetanus incidence rate of 7.8/100,000 (2019), mortality rate of 6/100,000 (2019) and under-five mortality of 41/1000 (2022). Agents were defined as under-five children, interacting with caretakers, health-worker and environment as vaccination and health facilities.

Results: According to Kenya Health Information System, there were 130 tetanus cases in April 2024; approximately, 60 cases of tetanus shall be averted; if we reduce zero-dose cases by 50%; through enhanced uptake of vaccinations targeting <70 tetanus cases monthly; by reducing probabilities of infection from 2 to 1.3 and of susceptibility-again-state from 2 to 3 months' post vaccination supported by ***SIS model*** calibration.

Conclusion: The model demonstrated that halving zero-dose vaccination reduces chance of tetanus cases and susceptibility by 2025.

Key words: zero-dose vaccination, agent based modelling, under-five, childhood immunization, tetanus

3. Introduction

Globally, vaccination is the cornerstone and a public health concern in the control of communicable diseases (WHO,2020). According to the WHO immunization goal; ninety (90%) of the children should be immunized (Cata-preta, 2021). Vaccination prevents 2.5 million deaths among the under-fives (U5) annually. One in five (1 in 5) children lacks access to vaccines in Sub-Saharan Africa (Ozigbu, 2022). Zero dose vaccination (ZD) refers to the lack of the first (1st) dose of diphtheria-tetanus-pertussis containing vaccine (DTP1) or the penta-dose that is DPT+ HiB & Hep B. Zero-dose is a global indicator among the under-fives. Zero-dose children remain a key objective and heart of the Immunization Agenda 2030 (IA2030) embedded in the strategy.

Statement of the problem

Globally, 14.2% under-five children are penta-zero dose, and 7.5% are truly zero dose (Chizoba, 2023). Globally, tetanus morbidity remains at 50,000 with an attributable mortality of 34,684 cases annually. In, Africa the tetanus incidence is 23,052 (World data, 2019)

In Sub Saharan Africa (SSA). The ZD remains at 7.7%, one vaccine-3.3%, two vaccines - 3.4%, three vaccines-14.6%, four vaccines- 70.9%, 59.9%- fully immunized (World Bank report 2022).

In Kenya ZD currently stands at 7%, indicating that seven (7) in 100 children under 1 year of age are ZD (KDHS,2022). Efforts are in place to improve the vaccination rate, but there remain access challenges in reaching all children essential vaccines from birth.

Currently the updates indicate that diphtheria and neonatal tetanus have been eliminated and pertussis is near elimination. There is also a marked control for the measles disease. This therefore leaves tetanus as an indicator of DPT, being the only disease yet to be eliminated or controlled among the DPT vaccination bracket (Our world in data 2019). Globally approx.- 50,000 cases

Tetanus cases in Kenya are currently 3338 as at 2019 and the targeted reduction of 50% means an approximate reduction to 1670 cases annually. Mortality due to tetanus stands at 2348 as at 2019 with an incidence rate of 7.8/100,000 and a mortality rate of 6/100,000 (World data 2019, KDHS, 2022)

For the purpose of this ZD project; tetanus shall be employed as an indicator for zero-dose. In Kenya, infant mortality rate (IMR) currently stands at 31.771/1000 live births (World data, 2019) and 41/1000lb for the under-five (U5) mortality rate.

Objectives of the project.

To determine number of tetanus cases that will be averted by reducing the prevalence of zero pentadose vaccination by 50% among under-fives by the year 2025.

Research question: How many tetanus cases will be averted if we reduce prevalence of zero dose vaccination by 50% among under-fives by the year 2025?

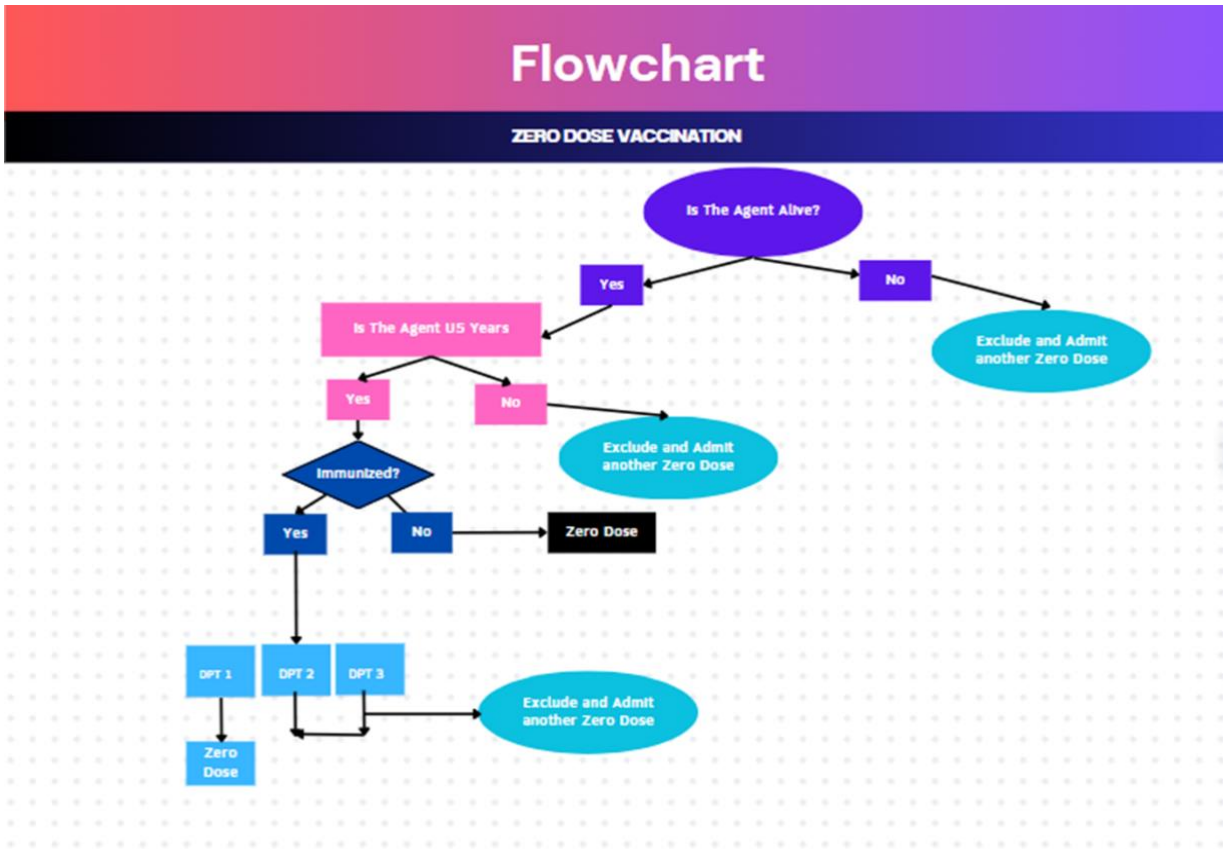
4. Methodology

The authors underwent a thorough training on agent based modelling and thereafter embarked on a zero dose vaccination model in Kenya.

The project targeted the Under-five children in Kenya, who live in the LMICs, specifically in Kenya. The main aim of the was to determine the number needed to vaccinate to achieve a 50% reduction in zero dose vaccination rates by 2025.

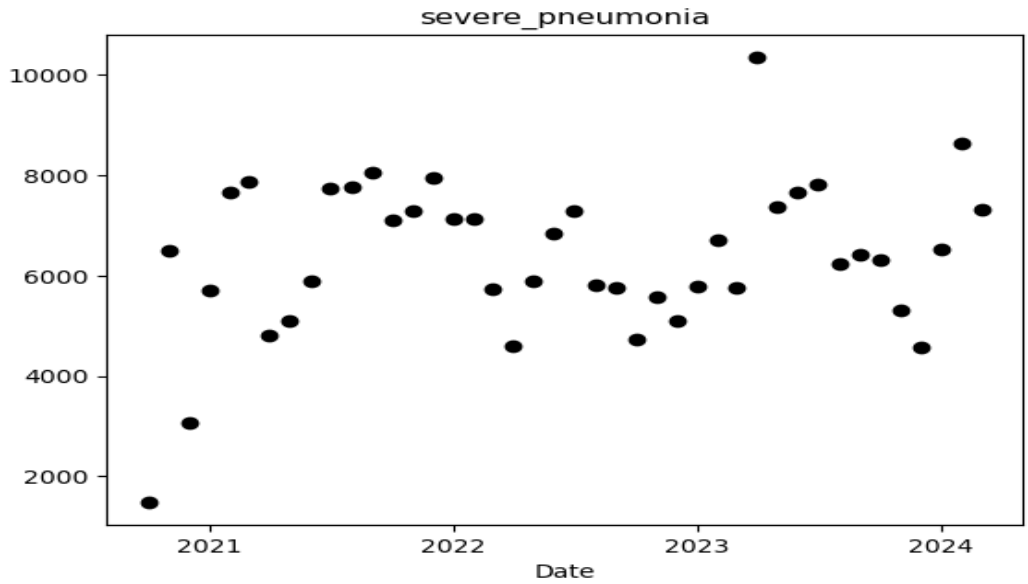
The outcome of interest being to reduce under-five morbidity and mortality rates through increased vaccination coverage by the year 2025. The project applied the Starsim - SIS model structure, with the agent being the under-five child who missed their first diphtheria, pertussis and tetanus vaccine here to referred as DPT 1. The perceived interaction are the under-five children caretakers, mother, guardian and the health care workers. The environment included in the project was the vaccination, the health facility and exposure to rusted metals or soil.

The inclusion and exclusion criteria is demonstrated in the flow chart below.

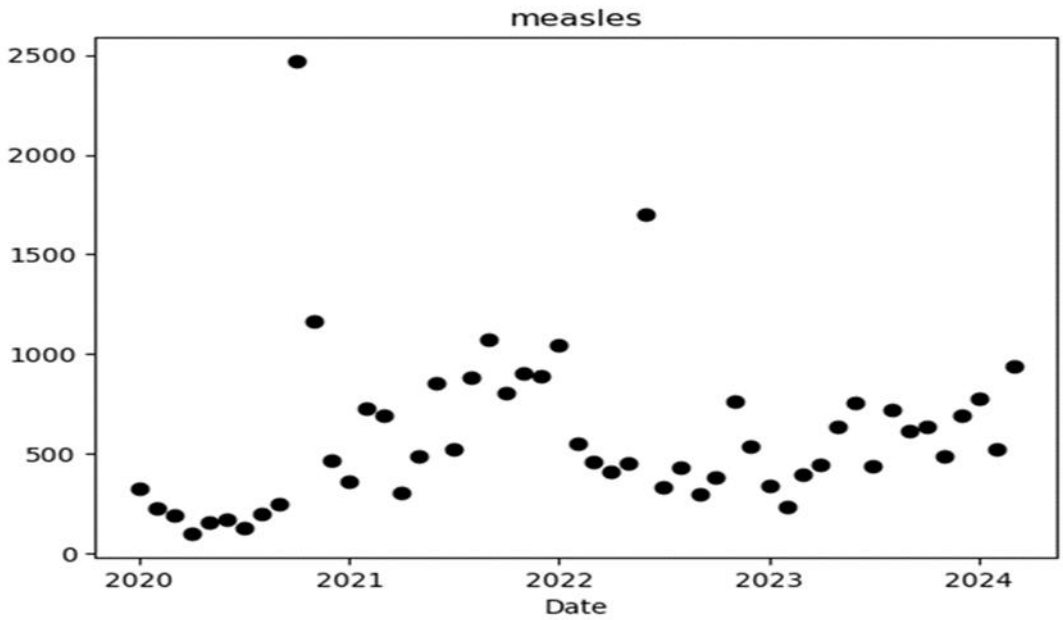


5. Results

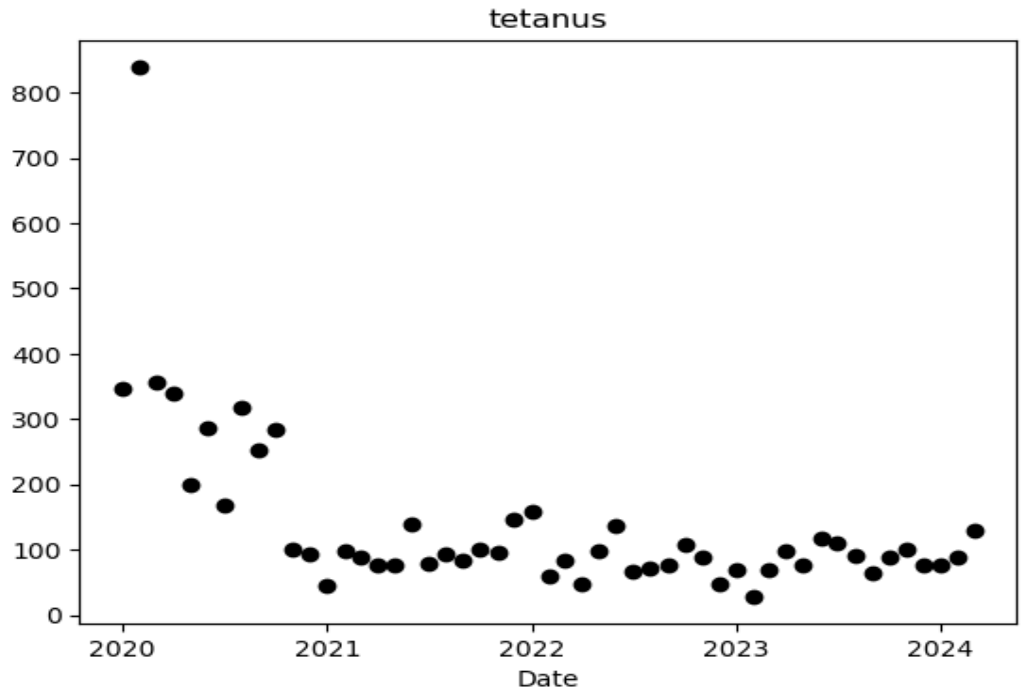
The tables below demonstrate the pneumonia, measles and tetanus cases from 2021-2024



There was an average of about 8000 cases of severe pneumonia monthly.



There was an average of approximately 600 measles cases monthly with exception of end of 2020 and mid 2022 that over 1000 cases.

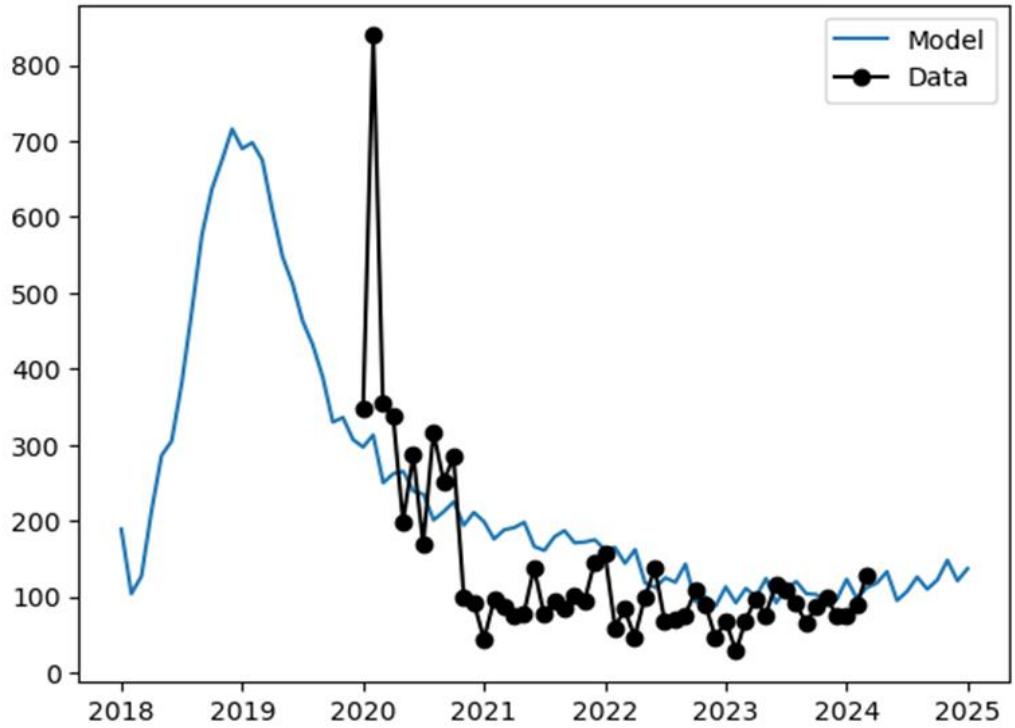


The government of Kenya was able to maintain tetanus cases at approximately 1000 monthly with exceptions of the year 2020 and 2022.

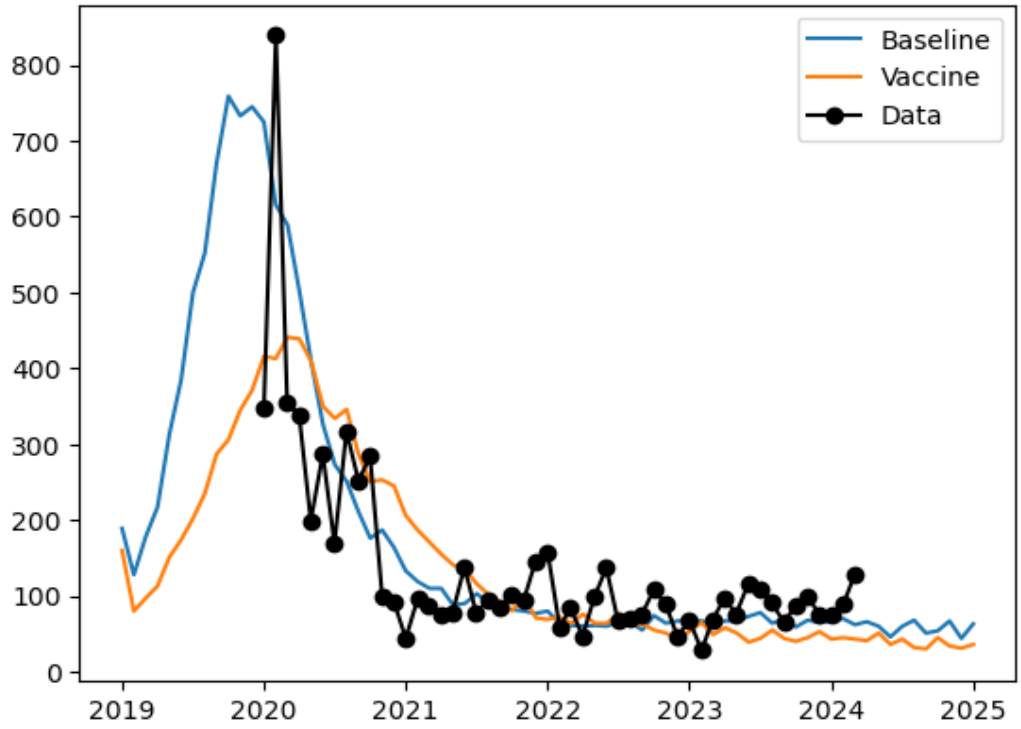
This was **an ABM structure starsim with susceptibility immunization and being susceptible again (SIS) model.**

The model was fitted with a Tetanus Beta=1.3, gamma =3/12, waning=0.055. The intervention applied was vaccination with a probability =0.25, efficacy=0.9 (vaccine to produce immunity). Tetanus cases averted by 50% approximate =75 cases/month

Model before calibration



Model after calibration



6. Discussion

The project proposed to achieve the 50% reduction in zero dose vaccination rates through introduction of vaccine campaigns. However, there was heterogeneity among the under-five children in Kenya. Equity in accessing vaccination remains a challenge especially DPT 1.

7. Conclusion

The model demonstrated that halving zero-dose vaccination cases from 3338 to 1670 will lead to a reduction in tetanus cases from 130 to 65 and susceptibility by 2025.

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